1) Bitwise shifting

```
#include <iostream>
using namespace std;
int main(){
int x;
   cout <<"Enter value: ";
   cin>>x;
   cout <<"Orignal value: "<<x<<endl;
   cout <<"After left shift(1): "<<(x<<1)<<endl;
   cout <<"After right shift(1): "<<(x>>1)<<endl;
   return 0;
}
```

2) C++ Program to Find Quotient and Remainder

output -:

```
#include <iostream>
using namespace std;
int main(){
int x,y;
   cout<<"Enter dividend: ";
   cin>>x;
   cout<<"Enter divisor: ";
   cin>>y;
   cout<<"Quotient = "<<x/y<<endl;
   cout<<"Remainder ="<<x%y<<endl;


   return 0;
}
```

3) Write a program to swap two numbers

output -:

```
#include <iostream>
using namespace std;
int main(){
int x,y;
   cout<<"Enter First number: ";
   cin>>x;
   cout<<"Enter Second number: ";
   cin>>y;
   cout<<"Before swapping:-"<<endl;
   cout<<"x="<<x<<"  "<<"y="<<y<<endl;
   cout<<"After swapping:-"<<endl;
   int temp;
   temp=x;
   x=y;
   y=temp;
    cout<<"x="<<x<<"  "<<"y="<<y<<endl;


   return 0;
}
```

4) Type casting & Type conversion

```
#include <iostream>
using namespace std;
int main(){
   // Type Conversion
  int x=10;
   float y=x;
//Type casting
  float a=12.13;
   int b=(int)a;
   cout<<"Type conversion: x="<<x<<" y="<<y<<endl;
   cout<<"Type casting: a="<<a<<" b="<<b<<endl;
   return 0;
}
```

Type casting :- char---→int

```
#include <iostream>
using namespace std;
int main(){
  char x;
cin>>x;
    cout<<(int)x;
    return 0;
}
```

Type casting:- int---→char

```
#include <iostream>
using namespace std;
int main(){
  int x;
cin>>x;
    cout<<(char)x;
    return 0;
}
```

Type casting int---→char x2

```
#include <iostream>
using namespace std;
int main(){
  int x;
cin>>x;
char ch=(char)x;
    cout<<ch;
    return 0;
}
```

conditional statements

IF ELSE Conditions

```
#include <iostream>
using namespace std;
int main(){
  int x;
    cout<<"Enter any digit: ";
  cin>>x;
    if(x%2==0){
        cout<<"Even Number";
    }
    else {
        cout<<"Odd Number";
    }
    return 0;
}
```

Take positive integer input and tell if is divisible by 5 or not.

```
#include <iostream>
using namespace std;
int main(){
  int x;
    cout<<"Enter any digit: ";
  cin>>x;
    if(x%5==0){
        cout<<"Yes It is divisible by 5";
    }
    else {
        cout<<"No it's not";
    }
    return 0;
}
```

Absolute Value

```
#include <iostream>
using namespace std;
int main(){
 int x;
   cout<<"Enter any digit: ";
 cin>>x;
   if(x<0) x=-x;
   cout<<x;

   return 0;
}
```

IF , IF ELSE, ELSE Statement (profit/loss)

```
#include <iostream>
using namespace std;
int main(){
 int cp,sp;
   cout<<"Enter cost Price: ";
 cin>>cp;
   cout<<"Enter Selling Price: ";
 cin>>sp;
   if(cp<sp){
      cout<<"Profit is: "<<sp-cp;
   }
   else if (cp>sp){
      cout<<"loss is: "<<cp-sp;
   }
   else {
      cout<<"No profit NO loss";
   }

   return 0;
}
```

Area of the rectangle is greater than its perimeter

```
#include <iostream>
using namespace std;
int main(){
 int l,b,area,peri;
   cout<<"Enter lenght: ";
 cin>>l;
   cout<<"Enter breadth: ";
 cin>>b;
area=l*b;
   peri=2*(l+b);
   if(area>peri){
      cout<<"Area is greater ";
   }
   else if(area==peri){
      cout<<"Both are equal ";
   }
   else {
      cout<<"perimeter is greater ";
   }
   return 0;
}
```

Multiple conditions statements using && and ||

Three digit number using (&&)

```
#include <iostream>
using namespace std;
int main(){
```

```
 int n;
  cout<<"Enter any digit: ";
 cin>>n;
  if(n>99 && n<1000){
     cout<<"Three digit number";
  }
  else {
   cout<<"Not A Three digit number";
  }
  return 0;
}
```

Divisible by 5 and 3

```
#include <iostream>
using namespace std;
int main(){
 int n;
  cout<<"Enter any digit: ";
 cin>>n;
  if(n%5==0 && n%3==0){
     cout<<"Divisible by 5 and 3";
  }
  else {
   cout<<"Not Divisible by 5 and 3";
  }
  return 0;
}
```

Sides of triangle

```
#include <iostream>
using namespace std;
int main(){
 int a,b,c;
 cout<<"Enter 1st side: ";
  cin>>a;
  cout<<"Enter 2nd side: ";
  cin>>b;
  cout<<"Enter 3rd side: ";
  cin>>c;
  if((a+b)>c && (a+c)>b && (b+c)>a){
     cout<<"Valid Triangle";}
    else {
     cout<<"Invalid Triangle";
    }
  return 0;
}
```

Greatest 3 Numbers of

```
#include <iostream>
using namespace std;
int main(){
 int a,b,c;
 cout<<"Enter First Number: ";
 cin>>a;
 cout<<"Enter Second Number: ";
 cin>>b;
 cout<<"Enter Third Number: ";
 cin>>c;
 if(a>b && a>c){
    cout<<a<<" is greatest";}
   else if (b>a && b>c) {
    cout<<b<<" is greatest";
   }
  else {
    cout<<c<<" is greatest";
  }
  return 0;
}
```

Least 3 Numbers of

```
#include <iostream>
```

```
using namespace std;
int main(){
 int a,b,c;
 cout<<"Enter First Number: ";
  cin>>a;
  cout<<"Enter Second Number: ";
  cin>>b;
  cout<<"Enter Third Number: ";
  cin>>c;
  if(a<b && a<c){
     cout<<a<<" is Smallest";}
    else if (b<a && b<c) {
     cout<<b<<" is Smallest";
    }
  else {
     cout<<c<<" is Smallest";
  }
  return 0;
}
```

Div. by 3/5 and not 15

```
#include <iostream>
using namespace std;
int main(){
  int n;
  cout<<"Enter any number(1-10)"<<endl;
cin>>n;
if((n%5==0 || n%3==0) && (n%15!=0)){
  cout<<"Divisible By 5 and 3";
}
  else{
     cout<<"Not Divisible By 5 and 3";
  }
  return 0;
}
```

Loops

While loop

```
#include <iostream>
using namespace std;
int main(){
  int n=0 ;
  while(++n <=5){
     cout<<"Hello"<<endl;
  }
}
```

While loopx2

```
#include <iostream>
using namespace std;
int main(){
  int n; //Initialization
  cout<<"Enter any number(1-10)"<<endl;
  cin>>n;
  while(n<=10){
     cout<<"value of number is:"<<n<<endl;
     n=++n;
  }
  return 0;
}
```

Do -While loop

```
#include <iostream>
using namespace std;
int main(){
  int n; //Initialization
    cout<<"Enter any number(1-10)"<<endl;
  cin>>n;
  do{
```

```
        cout<<"value of number is:"<<n<<endl;
    }
    while(n++ <=10);
    return 0;
}
```

For loop

```
#include <iostream>
using namespace std;
int main(){
     cout<<"Enter any number(1-10)"<<endl;
 for(int n=1; n<=5; n++){
     cout<<"Number is: "<<n<<endl;
 }
    return 0;
}
```

Nested IF ELSE

```
#include <iostream>
using namespace std;
int main(){
    int a,b,c;

    cout<<"Enter First number: ";
cin>>a;
    cout<<"Enter Second number: ";
cin>>b;
    cout<<"Enter Third number: ";
cin>>c;
  if(a>b){
      if(a>c){
          cout<<a<<" is Greatest";
      }
      else{
          cout<<c<<" is Greatest";
      }
  }
   else {
      if(b>c){
          cout<<b<<" is greatest";
      }
      else{
          cout<<c<<" is greatest";
      }
   }

    return 0;
}
```

IF ELSE Ladder

```
#include <iostream>
using namespace std;
int main(){
    int x;
    cout<<"Enter marks: ";
  cin>>x;
if(x>=81 && x<=100){
    cout<<"Very Good";
}
    else if (x>=61){
      cout<<"Good";
    }
     else if (x>=41){
      cout<<"Average";
    }
     else{
      cout<<"Fail";
    }
```

```
    return 0;
}
```

Ternary operator

```
#include <iostream>
using namespace std;
int main(){
  int x;
    cout<<"Enter any digit: ";
  cin>>x;
    // (condition) ? true : false ;
    (x%2==0) ? cout<<"Even": cout<<"Odd";

    return 0;
}
```

Calculator

```
#include <iostream>
using namespace std;
int main(){
    int a;
    cin>>a;
    char op;
    cin>>op;
    int b;
    cin>>b;
    if(op=='+') cout<<a+b;
    if(op=='-') cout<<a-b;
    if(op=='*') cout<<a*b;
    if(op=='/') cout<<a/b;
}
```

Calculator using Switch:-

```
#include <iostream>
using namespace std;
int main(){
    int a;
    cin>>a;
    char op;
    cin>>op;
    int b;
    cin>>b;
  switch(op){
    case '+':
        cout<<a+b;
        break;
    case '-':
        cout<<a-b;
        break;
    case '*':
        cout<<a*b;
        break;
    case '/':
        cout<<a/b;
        break;
    default:
    cout<<"Invalid operator";
  }
}
```

